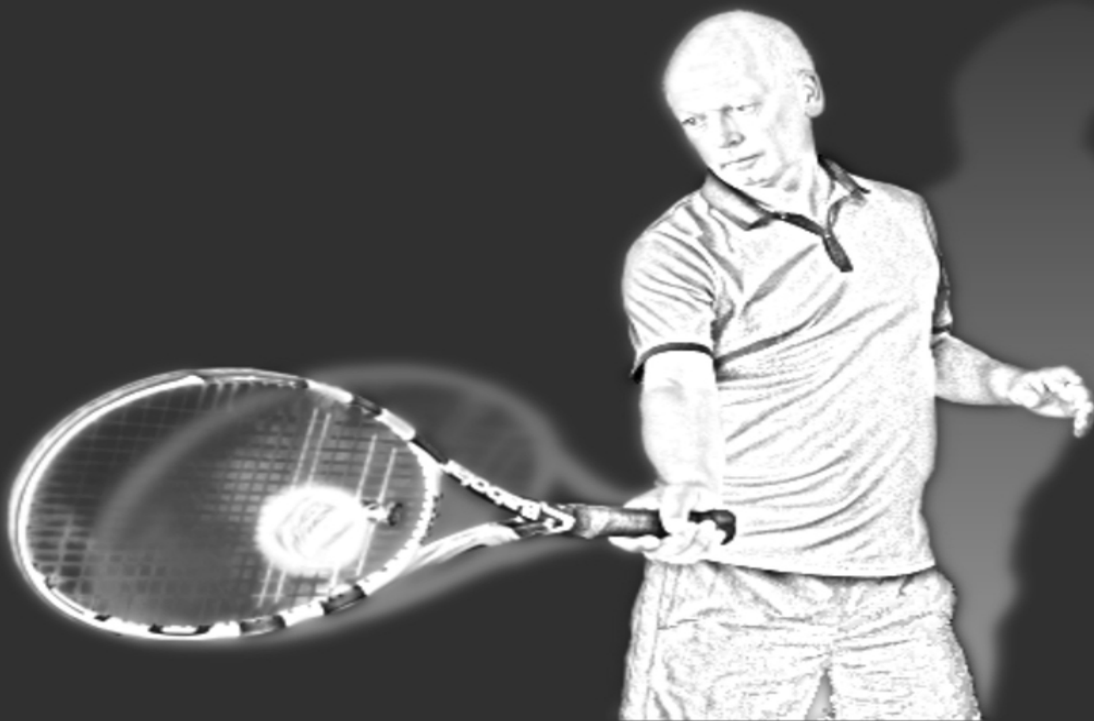


A must read - How to improve your forehand today.
For players of all ages and levels

one minute tennis FOREHAND SOLUTION



Improve Faster - Play Smarter

STEPHEN JAMES BOURNE

one minute tennis
**FOREHAND
SOLUTION**

STEPHEN JAMES BOURNE

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The One Minute System

"Hi, I'm Steve, welcome to my One Minute Tennis System."

For many years I was privileged to run a very successful tennis academy. I worked with high-level players, world-class juniors and recreational players of all ages and levels.

During this time I studied anatomy and biomechanics and developed a deep understanding of the scientific principles of tennis.

Utilizing my knowledge of the science and physics of the game I created a unique teaching system based on feel, sensation, and imagery. This system proved to be incredibly successful with players of all levels improving faster than they ever thought would be possible.

The Language of the Mind is Words- The Language of the Body is Feeling!

The One Minute System is based on a simple principle.

Essential teaching - the moment where the player understands and learns a new movement is much more effective if it occurs in One Minute or less!

Traditional teaching methods label us as either a

Visual Learner

Audible Learner

Kinaesthetic Learner

This is often called VAK teaching and is how tennis has been taught throughout history.

People often say

"I have to see this written down"

"I have to see things" or

"I have to feel this"

Of course, they are right, seeing, feeling, and touching are essential for learning practical skills.

But ... I created a better way, helping players to learn incredibly quickly, in an almost superhuman fashion. To achieve this we communicate to all of the learning senses at the same time.

This is called *Blended Learning or Blended Teaching*.

1. Feel the movement,



2. See the movement



3. Understand the movement.



How to Use this Guide

There are many factors that make the difference between winning and losing in tennis. Fitness, athletic ability, insufficient practice time, technique, and mental strength under pressure.

For the vast majority of players, **the biggest limitation is 'technique'.**

One Minute Tennis Solutions concentrate on the technical aspect of the game and are beneficial for players of all levels.

I have included detailed information on the biomechanical principles of each stroke. Accompanied by simple and effective 'One Minute Solutions'

Science helps you to understand the stroke

One Minute Solution helps you improve your stroke.

At the end of each One Minute Solution, there is a link to the same lesson in video format

These solutions work almost instantly.

To benefit from the advice, take your time to feel and visualize the shapes and movements.

At first use shadow strokes and slow controlled practice and then you will see your real tennis potential emerge.

As well as using the One Minute Solutions you can also study the precise science and biomechanics of each aspect of the strokes to understand how the modern tennis game really works.

Visualize, Feel & Understand Tennis

Forehand Take Back Biomechanics

The biomechanics of tennis refer to the study of the movement and function of the human body during the execution of tennis strokes. This includes the analysis of factors such as joint angles, muscle activation patterns, and kinetic and kinematic variables.

Understanding the biomechanics of tennis can help coaches and players optimize technique and improve performance.

There are several key principles to consider when analyzing the biomechanics of tennis strokes:

1. *Force production*: The ability to generate power and speed in a stroke is important for hitting shots with depth and pace. This is influenced by factors such as muscle strength and coordination, as well as body position and alignment.

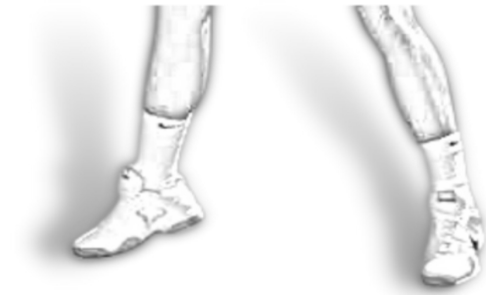
2. *Movement patterns*: The way in which a player moves around the court and sets up for shots can impact their effectiveness. This includes factors such as footwork, balance, and body positioning.

3. *Kinetic chain*: The kinetic chain refers to the sequential activation of muscles and joints from the ground up, starting with the feet and ending with the hand or racquet. Optimizing the kinetic chain can help players generate more power and control.

4. *Joint angles*: The angle at which joints are positioned can impact the efficiency and effectiveness of a stroke. For example, a player with a high elbow position in their backhand may have difficulty generating power and control. By analyzing these and other factors, coaches and players can identify areas for improvement and make changes to the technique in order to optimize performance.

Initial Unit Turn

Effective preparation starts as soon as the opponent hits the ball, usually after landing the split step with a unitary body turn involving the feet, torso, and racket.



The common and traditional advice to immediately "take your racket back," with the independent motion of the arm is flawed. Avoid doing it.

Start unit turn (the body from the waist to the neck) and turn immediately following the split step.

The shoulders initially turn at an angle of about 45 degrees. The takeback involves a combination of movements at the shoulder, elbow, wrist, and fingers.

The shoulder rotates and the elbow flexes to bring the racquet back behind the body. The wrist pronates (rotates inward) to prepare for the impact with the ball.

The fingers and hand also play a role in gripping and positioning the racquet during the takeback.



The opposite arm may be straight or slightly bent, the hands may separate at slightly different intervals and rotations of the shoulder, hip, and legs are crucial.

Although the hands may start to rise slightly and the angle of the racket's face may change slightly, the opposite hand remains on the racket and it moves naturally with the body turn. *Most often, these variations are personal.* The racket is still in front of the torso in this initial phase and is only moving back naturally as a result of the body's rotation.

The takeback of the racket engages a number of muscle groups in the shoulder, arm, and hand.

The rotator cuff muscles (such as the supraspinatus and infraspinatus) help to rotate the shoulder.

The elbow flexors (such as the biceps brachii and brachialis) allow the elbow to bend.

The wrist pronators (such as the pronator teres and pronator quadratus) enable the wrist to rotate inward.

The hand muscles (such as the flexor digitorum and extensor digitorum) assist with gripping and positioning the racquet.

Second Phase of the Turn

The hands gradually separate as the turn progresses, and the left arm and hand move to point across the body, nearly perpendicular to the sideline until the shoulders are at least 90 degrees from the net, or even further, the hips and shoulders continue to rotate.

*Many athletes, such as Djokovic, hold on to their opposing hand longer, **but the left arm being parallel to the net is the pre-requisite.***

A little later and with a partial straightening of the arm, the left arm stretch begins. The straight arm and possibly a slight delay in the separation of the hands, however, will help the majority of players get a better turn.

The weight is shifting to the outside right leg as the turn is completed, and the knees are coiling. The left heel is typically partially elevated and the left leg's weight is primarily distributed to the foot. With the spine almost at a 90-degree angle to the court and only a slight tilt to the right, the posture is upright.

Backswing

When the turn is finished the hand moves upward on a diagonal and slightly outside or away from the body, reaching shoulder height or just below.

As the hand, arm, and racket move downward in preparation for the forward swing from the outside backswing position, they rotate backward from the shoulder.

The shoulder joint is the main joint involved in the take-back phase of the forehand stroke. This joint is a ball and socket joint that allows for a wide range of motion, including flexion, extension, abduction, and adduction. The rotator cuff muscles, which include the



supraspinatus, infraspinatus, teres minor, and subscapularis, help to stabilize the shoulder joint and allow for precise control of the racket.

The racket descends from this backswing position, **the hitting hand remaining on the player's right side.** The hitting arm and racket move down and externally rotate backward to position the hitting arm for the forward swing at the top of the backswing. In the movement toward contact, this position prepares the racket arm for maximum shoulder acceleration.

There is considerable variation in the backswings of the top players, as an example of this, Djokovic and Nadal's racket movement, racket tip angle, and backward arm rotation are slightly different.

There are various hand heights, racket tip heights, racket tip angles, shapes in the racket paths, and separations from the torso, among other things.

The same is true for other top players. When Djokovic's racket face is parallel to the baseline after tilting the tip forward, he continues to rotate his external arm backward. Nadal raises the racket with its face slightly closed and, without using any forward tilt, points the tip backward. These differences only highlight the importance of the five fundamental commonalities.

When there are so many variations. It is essential to get the recognized common fundamentals right!



The racket hand should stay on the right side of the body.

The grips can appear to affect this fundamental as players with more extreme grips often draw the racket behind the body

Even though the racket tip is angled behind the torso, extreme grips allow the hand to be on the hitting side.

So depending on the grip, the racket and the hand, or only the hand will remain on the right side of the body.

The fundamental is that the hand is always on the right side of the body, irrespective of the grip.

Stance

There is a lot of misunderstanding regarding stances and how they apply at the lower levels. "Play open stance like the pros is one adage you hear."

Typically, this is understood to mean that the feet should be parallel to the baseline and the stance should be completely open. *Amazingly, knowledgeable tour trainers also share this advice in addition to online gurus.*

In actuality, pro players use semi-open stances for the vast majority of all forehands. The feet are not parallel to the baseline and a line across the feet will typically point 30 to 45 degrees off the baseline.

A fully open stance lessens coiling and is fantastic for redirecting power, but not for generating power.

With a fully open stance, the rotation of the hips, left leg, and left foot is limited if the feet remain parallel to the baseline.

Coil and load capacity are limited for players.

The fundamental commonality is that the player loads the weight into the right foot and the right side of the body!

Forward Torso Rotation

Modern forehands require a semi-open stance in addition to coiling during the pre-swing phase.

This is because during the forward swing, the shoulders can rotate up to 180 degrees, and occasionally even a little bit more.

It is typical for the rear shoulder to turn around until it is facing the net from the full coil with the shoulders facing the net. This can happen naturally thanks to the left foot's semi-open alignment.

During the forward swing, the back shoulder turns 180 degrees and ends up facing the net.

So semi-open allows for maximum uncoiling and maximum coiling, respectively. Forehand speeds up to 100 mph or even higher and spin up to and exceeding 3000 rpm have increased, and these two factors are the main causes of these increases.



Forehand Take Back Solution

"Hold onto the ball"

As seen in the biomechanical analysis there are several common denominators that apply to every professional player and are essential for consistent and accurate forehand drives. For our instant solution to be truly successful all of these reference points must be met!

1. The racket must be taken back with the elbow first.

Not the racket first.

If the racket leads then the weaker, lateral side of the arm is going to begin the upswing to the ball.

When the elbow goes back first then the stronger medial side of the arm is going to pull the racket into contact.



2. The weight must load onto the outside foot.

This encourages rotation and forwards momentum.

This can be a 'Semi-Open Stance (as illustrated) Open or Neutral Stance.

But the weight will always load on the outside.



3. The left arm must achieve a position that is parallel to the net. This can be achieved by keeping hold of the racket with the left hand until the end of the take back and then moving the arm forwards to a position that is parallel to the net.
Or by simply placing the hand into this parallel position.

It is the rotation of the arm that is essential.
It should be a strong and definite movement ensuring that the stroke has a clear and precise point to begin from every time.



4. The shoulders should rotate to about 65° and be more or less facing away from the intended target of the stroke.

This movement should be blended with the positioning of the arm parallel to the net.

It is important to be aware of the correct shoulder rotation.

A very common cause of inconsistency in the Forehand drive is a lack of or incorrect shoulder turn.

5. The head should slightly lead onto the right side of the body.

Remember, the body will follow the head.

So if the head is positioned onto the right side of the body, then the weight will naturally lean in this direction, and therefore when the stroke begins, the weight will move from right to left creating rotational energy and angular momentum.

These are the five essential references required for a perfect Forehand take back. Obviously, this is a lot to learn, for many players by the time they have remembered the 5th reference point, they have forgotten the first.

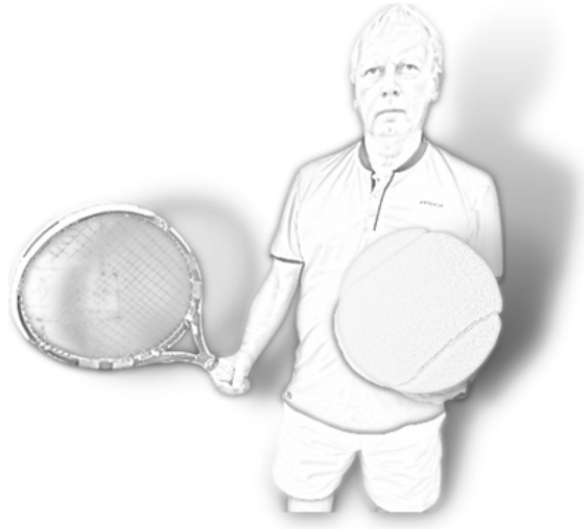
Lots of repetition is required to create a correct muscle memory of all five movements.

"the One Minute Solution is a better way"

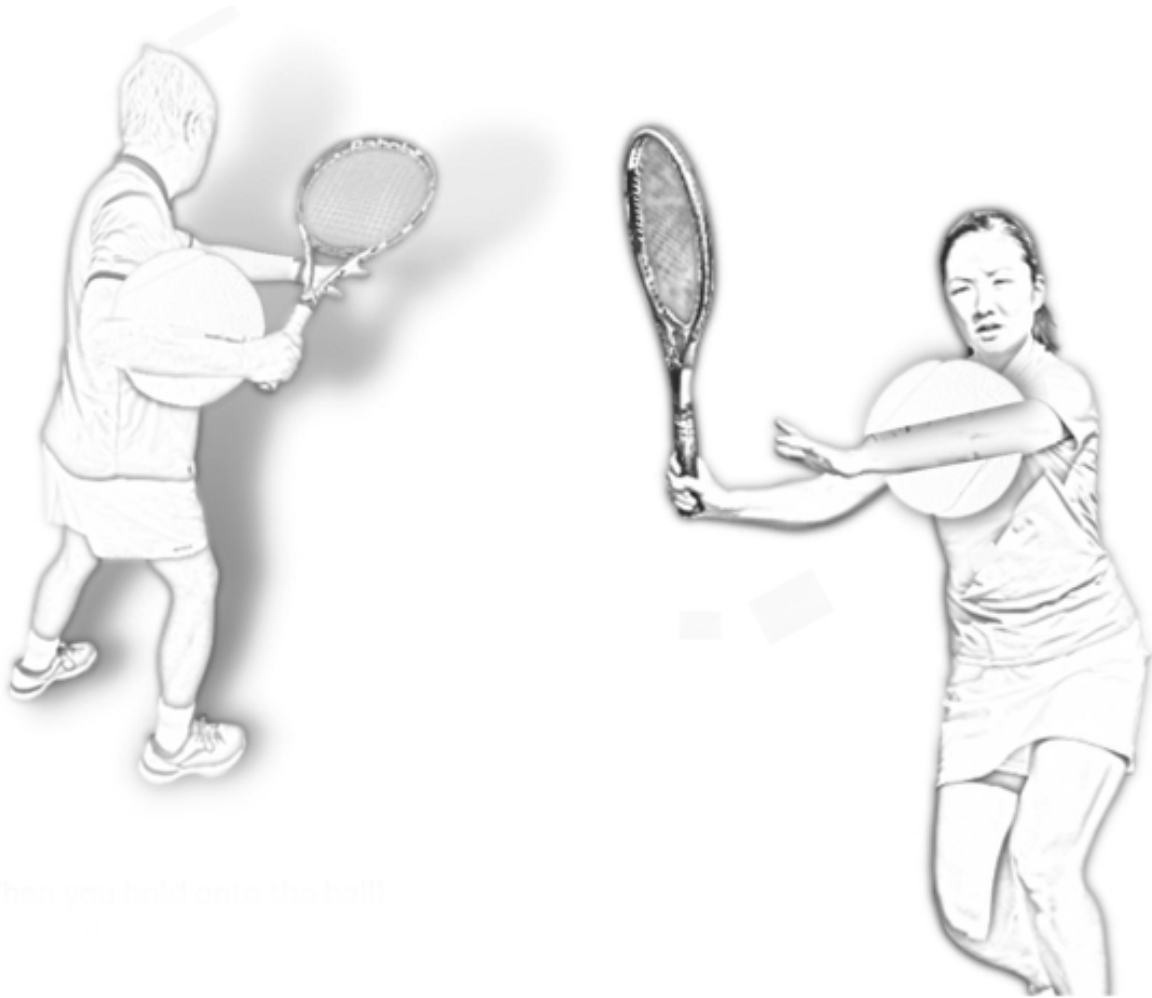


The One Minute Solution

As you begin the forehand stroke, **simply imagine that you have a large ball nestled into your arms.** When you take the racket back, make sure that you continue to cradle the imaginary ball. **With this imaginary ball in your arms the Forehand Take Back will be perfect**



1. *(When you hold onto the ball)* **The racket will naturally go back with your elbow first.** Any other movement and the imaginary ball would fall.



(When you hold onto the ball)

your weight will automatically happen.

2. (When you hold onto the ball) **Your weight will automatically load onto the outside foot.** This requires no thinking - it just happens.

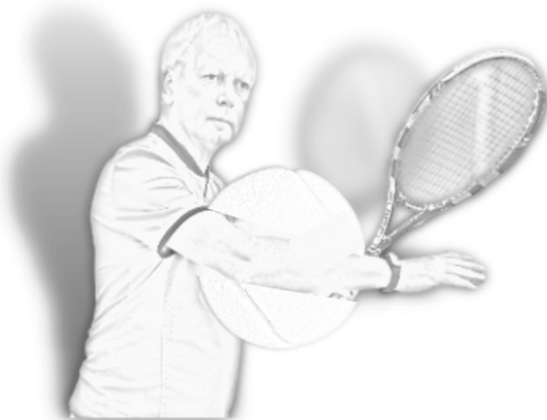


3.(When you hold onto the ball)

Your left arm cannot avoid being parallel to the net.

4. (When you hold onto the ball)

your shoulders have to rotate away from the target.



5.(When you hold onto the ball)

Your head cannot avoid leaning over to the right side of the body.

By applying this simple, easy to understand, easy to apply, easy to remember picture of **"holding onto an imaginary large ball"** then the Forehand take back will be perfect every single time.



By using this image, the perfect Forehand take-back becomes incredibly easy.

The movement becomes natural in minutes and if you feel that they are losing the correct shape and reverting to bad habits then even while playing, you can picture the ball in your arms and correct the technique.

For a perfect, consistent, and accurate Forehand take back

"Hold onto the Ball"



THE LANGUAGE OF THE MIND IS WORDS

THE LANGUAGE OF THE BODY IS FEELING

To see this lesson on YouTube
[https://www.youtube.com/watch?
v=fjHUreX5nxA&t=6s](https://www.youtube.com/watch?v=fjHUreX5nxA&t=6s)

Forehand Swing Path

Biomechanics

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1. Force production: The ability to generate power and speed in a stroke is important for hitting shots with depth and pace. This is influenced by factors such as muscle strength and coordination, as well as body position and alignment.

2. Movement patterns: The way in which a player moves around the court and sets up for shots can impact their effectiveness. This includes factors such as footwork, balance, and body positioning.

3. Kinetic chain: The kinetic chain refers to the sequential activation of muscles and joints from the ground up, starting with the feet and ending with the hand or racquet. Optimizing the kinetic chain can help players generate more power and control.

4. Joint angles: The angle at which joints are positioned can impact the efficiency and effectiveness of a stroke. For

example, a player with a high elbow position in their backhand may have difficulty generating power and control. By analyzing these and other factors, coaches and players can identify areas for improvement and make changes to technique in order to optimize performance.

Top of the Backswing

The player positions the hitting arm as a unit preparing for the racket, hand, and arm to descend.

The hitting arm shapes can be divided into two basic categories: bent elbow and straight elbow.



An elbow that is bent inward toward the torso is said to be bent.

The grips, the length of the player's arm, and preference all have some influence on the precise shape, angle of the bend, and distance from the body.

Novak Djokovic, Stan Wawrinka, and Jack Sock are a few of the many players who used bent elbow forehands. Wawrinka's grip is more relaxed with less bend, Novak's is more, and Sock's is the tightest with his hand completely encircling the handle.

The elbow straightens out with a straight arm, creating a roughly straight line from the shoulder to the hand. Rafael Nadal, Leyla Fernandez, Grigor Dimitrov, and Roger Federer all play in this set-up.

Height of Hand and Racket.

The next reference is where the hitting arm, hand, and racket are in relation to the height of the approaching ball.

A common misconception is that it is essential to drop the hand "well" below the level of the ball so that you can swing "low to high."

The hand may, at most, fall a few inches below the ball. However, on occasion, it never actually descends below it.



The hand is secondary in importance and *the height of the racket head is the most important factor.*

Sometimes, dropping the racket head so that the tip points downwardly at the court is caused by the arm rotating backward rather than the hand. When players use this backward rotation, the racket head may be anything from a few inches to a foot below the ball.

However, this only occurs when players employ more extreme variations of the so-called windshield wiper—they rotate the arm and racket backward from the shoulder to set up additional forward rotation on the way to contact and then out into the follow-through.

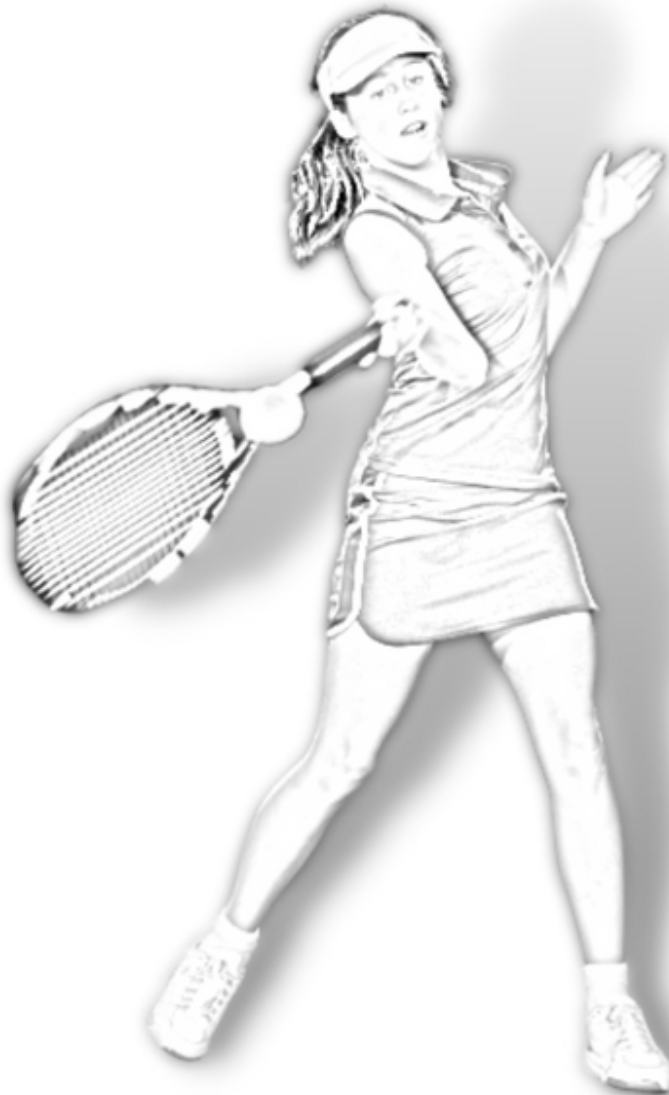
You don't want to artificially drop the hand and alter the hitting arm shape to hit "low to high" for this reason. You don't actually need to think about it.

To hit up and out to the extension position on a powerful drive, simply pulling the hand forward is probably the best course of action from the outside backswing position.

Upswing

From a biomechanical perspective, the forehand swing involves a combination of rotational and translational movements.

The rotational movements occur around the player's spine, with the hips and shoulders turning as the swing progresses. The translational movements involve the shifting of weight from the back foot to the front foot, as well as the forward movement of the racquet as it makes contact with the ball.



In terms of anatomy, the forehand swing relies on the muscles of the shoulder, upper back, and core to generate power and control the racquet. The muscles of the lower body, such as the quadriceps, glutes, and calves, also play a role in helping the player maintain balance and stability during the swing.

Three forward motions are made by the hand and racket: **upward, outward, and also across**. These are three clearly and distinctly different dimensions.

The hand and racket start close to the body and move outward to the player's right before moving inside out toward the contact.

The hand's motion and forward rotation propels the racket upward at the same time.

At the point of contact, the curve of the swing changes, and the hand and racket start to cross the player's body from the right to the left.

The racket is still pointing upward, outward, or in the opponent's direction. Once more, a curve that changes direction three times.



The forward swing begins when the hand roughly reaches eye level and the left edge of the torso at the same time.

The idea that emphasizing the hand and racket's motion to the player's left across the body is another message that is frequently found online. "Bend the elbow sharply and pull the racket across before making contact". or a variation on that theme.

This is false. The fundamental shape of the hitting arm structure is maintained well into the forward swing and out toward the extension point.

Pronation

The hand, arm, and racket are almost always rotated or pronated to some extent when the top tennis players reach the extension point on a drive.

When this rotation is optimized, the racket tip may be directly pointed at the sideline, to the left.

The swing has also expanded in other key areas, like the large space between the torso and the hand.

In the so-called classical forehand, the racket remains primarily "on edge" through the extension while pointing upward or forward. To find professional athletes who utilized this less complex forward swing pattern, you must look back to the era of players like Pete Sampras, Tim Henman, or Andre Agassi.



The rise of the universal pronated and rotational swing path along with polyester strings is responsible for the astronomical spin levels in professional tennis.

Players who used more traditional forward swings generated only about 2/3 as much forehand spin, or 2000 rpm maximum as opposed to 3000 rpm on average in today's game.

The source of this huge increase in spin and speed is clearly the swing path and not the extreme grips.

Juan Martin Del Potro and Roger Federer prove this with slightly modified eastern grips yet a full range of pronated swing paths to some extent on virtually every ball. They hit with enormous power and spin and their more "old-fashioned" grips do not affect their performance in any way.

There is enormous variation in the forward swing depending on the height, speed and spin of the ball.

The elbow and hand can and frequently do come across more quickly.

With the hand, lower, shorter finishes are feasible. It's possible for the pronated action to be stronger or softer, as well as to move more quickly or more slowly. The elbow bends include more, less, sooner or later, and others.

With shorter balls, deeper balls, wider balls, and lower balls, these factors come into play. Also when the player is trying to hit with a specific shot shape—more arcing, deeper more angled, shorter angled, with more or less spin and more or less speed.



The difficulty in analyzing the stroke occurs at the finish, the last stage of racket deceleration. If you hit a forehand at 90 mph, your racket is probably traveling at around 60 mph at the point of contact. It is traveling at around 5 mph when it reaches the end of the swing.

If you swing to the extension and hit the checkpoints, you will obviously slow down on the follow-through. The speed of this deceleration will affect where the racket finishes.

Excellent players extend and accelerate on powerful drives. This outward extension results in the fastest racket head speed at the point of contact.

Many recreational players miss this acceleration and are either neutral or even decelerating towards and through contact with the ball.

Despite a lot of information to the contrary. **There is no "wrist snap" You cannot snap the wrist forward if it is not relaxed. In fact, Nadal keeps his wrist angle almost constant throughout the entire forward swing.** His extreme pronation is not a "wrist snap" it is created by the extreme upward and outward motion generating a ton of power and spin.

Contact



Contact is the crucial turning point.

The exact moment of contact happens in the middle of the whole swing and almost always occurs when the torso is halfway through the rotation.

Professional tennis players have shoulders that can rotate 180 degrees, allowing them to swing forward and rotate at the same time. At the point of contact, the torso is right in the middle of this rotation.

The total forward rotation, especially when using neutral stances, is about 90 degrees. This shows that the shoulders are at a 45-degree angle in relation to the baseline at contact with the ball.

Concerning the body, it is clear that the hand and racket are to the side and forward. These distances, however, depend on the hitting arm style and happen naturally and automatically.

If the player uses the straight arm forehand then contact is typically slightly more forward and further away from the body. bent arm, slightly less out in front and a little bit closer. Understanding these connections is primarily academic. Again, the precise position is determined by the shape of the hitting arm and the upward, outward, and cross-motions of the forward swing.

If you have the preparation and extension right, then the contact point will also be right so concentrating on the preparation and spacing to the ball is the priority.

Forehand Swing Path Lesson

One Minute Solution - PowerPoint

Most players will naturally slow the racket down when the points become important and the pressure is really on. Also, many players struggle to hit with speed and topspin in any situation.

The instant solution for this must encourage the correct swing path of the racket accompanied by confident, faster racket speed

The answer to this problem is the swing path of the racket. Remember the swing path of the racket dictates the shape of the ball.

However to create real topspin and swing like the pro players is an extremely unnatural action.

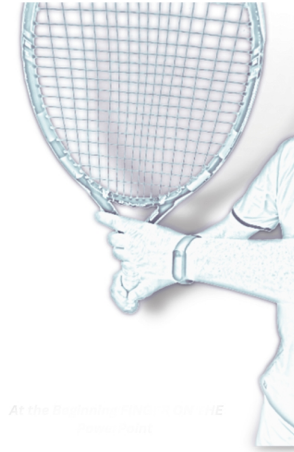
Any instant solution must assist both the swing path and the speed of the racket. An awesome way of achieving this is by using the **Power Point**



Place something like a piece of tape or play-doe onto the Throat of the racket. This we are going to call, the **power point!**

Now you have something that you can physically feel on the throat of your racket. You have a "tactile reference."

In a game played at the speed of tennis. *"Tactile references" are incredibly more efficient than any other method.*



At the Beginning FINGER ON THE PowerPoint

As you begin in the ready position the **PowerPoint** should be on the bottom edge of the racket, facing down and therefore the finger of your left hand will naturally be resting on the **PowerPoint**.



Take the racket back with the left hand and the finger remaining on the power point as long as feels comfortable.

The result of this will be a correct unit turn and the weight of the body loading into the right side.

Now, swing hit the ball and continue the stroke until the left hand collects/catches the racket.

The thumb must catch and connect with the PowerPoint.

*It will take a few goes as your natural movement will be, to begin with, your finger on the **PowerPoint** and end with your finger on the PowerPoint.*

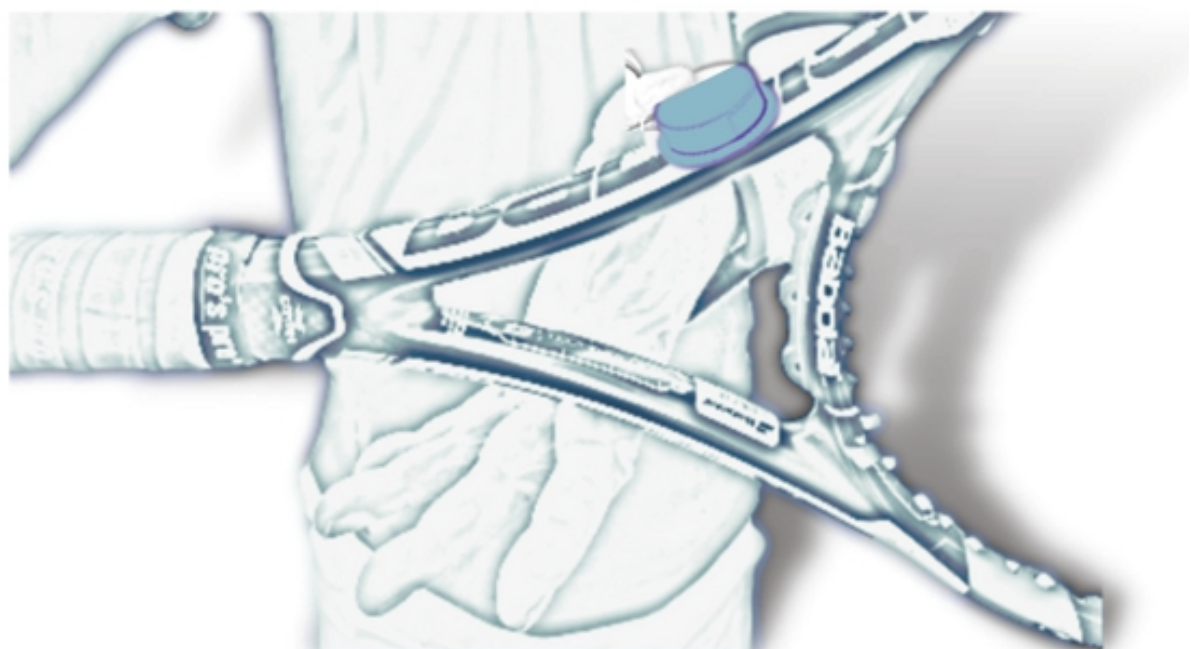
Persevere until you can feel your finger-to-thumb movement.

That is your finger on the **PowerPoint** at the beginning of the stroke and your thumb on the **PowerPoint** at the of the stroke



At the End THUMB ON THE PowerPoint

Make several Shadow swings.
Practice with your eyes closed, until you have the swing correct every time.



***THE LANGUAGE OF THE MIND IS WORDS
THE LANGUAGE OF THE BODY IS FEELING***

Now, when you hit the ball with your finger on the **PowerPoint** at the beginning of the stroke and your thumb on the **PowerPoint** at the end of the stroke.

Your Forehand will be faster and have much more spin.

The finger-to-thumb and use of **PowerPoint** create a wonderful paradox. **The harder you hit your new Forehand, the more the ball goes in.**



With this swing path, if you increase the ball speed by 10% the spin increases by 15%

If you increase the ball speed by 40% the spin increases by 60%

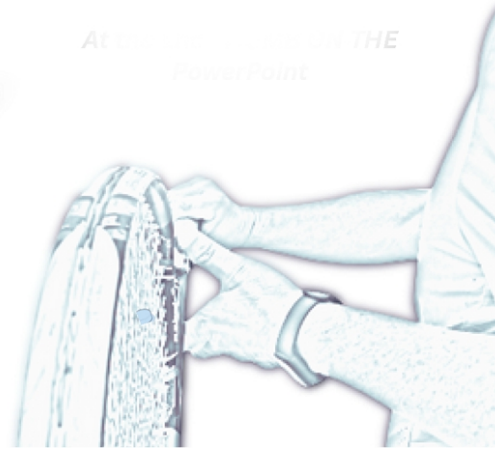
By using a stroke where swinging faster creates more spin than speed you will be able to hit faster yet safer than ever before.

Use the power point and hit the ball like you could not believe was possible.



***At the Beginning FINGER ON THE
PowerPoint***

***At the End THUMB ON THE
PowerPoint***



To see this lesson on YouTube
[https://www.youtube.com/watch?
v=C9O2GZkPcwU&list=PL0bA4JdHfGN78sdJWrw9B7__sEenkIIH
Z&index=12&t=176s](https://www.youtube.com/watch?v=C9O2GZkPcwU&list=PL0bA4JdHfGN78sdJWrw9B7__sEenkIIHZ&index=12&t=176s)

Forehand Wrist Position

Biomechanics

The biomechanics of hitting a tennis forehand involves the use of several joints and muscles in the hand, wrist, and arm to generate power and control the direction of the stroke.

The hand and wrist play a key role in the forehand stroke. *The wrist is responsible for the rotation of the racquet, which generates spin on the ball. The wrist also helps to absorb shock and reduce the risk of injury to the elbow joint.*

The muscles of the hand and wrist also contribute to the power and control of the stroke.

The forearm muscles, including the wrist flexors and extensors, are responsible for the wrist's rotation and the racquet's movement.

The muscles of the hand, such as the flexor and extensor digitorum, help to grip the racquet and control the direction of the stroke.

Overall, the forehand stroke requires the coordinated use of several joints and muscles in the hand, wrist, and arm to generate power and control the direction of the ball.

The muscles, ligaments, and tendons that are involved in hitting a tennis forehand include:

Wrist flexors: These muscles, located in the front of the forearm, are responsible for bending the wrist and rotating the racquet. The flexor carpi radialis and flexor carpi ulnaris are the main wrist flexors.



- Wrist extensors: These muscles, located in the back of the forearm, are responsible for straightening the wrist and rotating the racquet. The extensor carpi radialis and extensor carpi ulnaris are the main wrist extensors.
- Forearm muscles: The muscles of the forearm, including the pronator teres and supinator, help to rotate the wrist and control the racquet.
- Hand muscles: The muscles of the hand, such as the flexor and extensor digitorum, help to grip the racquet and control the direction of the stroke.

Ligaments:

- Wrist ligaments: The wrist is held together by a network of ligaments that connect the bones of the hand and wrist. These ligaments provide stability and support to the wrist joint.

Tendons:

- Tendons of the wrist and hand: The tendons of the wrist and hand, including the extensor tendons and flexor tendons, attach the muscles of the forearm to the bones of the hand and wrist. These tendons help to move the hand and wrist and control the racquet.



Overall, the hand and arm hitting a tennis forehand involve the coordinated use of several muscles, ligaments, and tendons to generate power and control the direction of the ball.

The problems occur when the grip of the racket or flexion of the wrist is too tight. This totally natural 'tight' feeling restricts the range of motion of the racket and reduces racket speed, power and control.

In the area of a wrist watch there are 32 muscles and 27 bones. When the muscles are flexed then the movement and flexion of the wrist is greatly restricted.

This problem of being too tight can be constantly present (in many recreational players) or simply occur as the pressure and importance of the point increases.

Forehand Wrist Position Lesson

The natural way of holding a tennis racket is with a tight and strong grip.

However, this type of tense, muscular contact with the racket prohibits a real feel for the ball. It prohibits real power. It prohibits good tennis.

99.9% of tennis players never enjoy the loose, free strokes of the best players in the world because they cannot experience the loose relaxed grip of the pro players.

The best players in the world use BOTH a loose grip and a loose wrist to generate speed and power. Unless this feeling is acquired at a very early age it is extremely unnatural and hard to utilize.

THE BOUNCE

A fantastic way of reproducing the loose relaxed feel required for world-class hitting is to find the bounce.

HOW TO FIND THE BOUNCE

Begin in the ready position with a relaxed and comfortable grip.

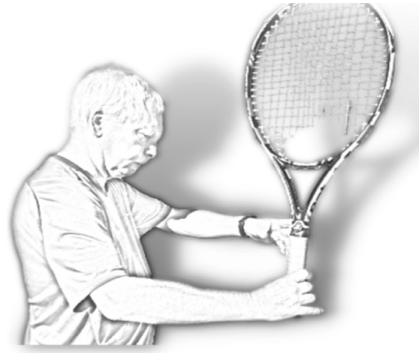
Take the racket back by rotating the body in a 'Unit Turn' Depending on the style of your stroke the racket will now be pointing to the sky or somewhere upwards and forwards at an angle of about 45 degrees.



Loose grip with the hand slightly open

Pause in this position and open your hand slightly. Now the racket grip will be smaller than the space in your hand and will almost fall through your fingers. If the racket begins to fall, then the space is too large.

Move your hand slowly backward and forwards (only a short distance) and if your grip is still larger than the handle of the racket you should feel the racket moving within the semi-open grip of your hand. If you feel your grip tightening on the racket.



Repeat the process until you are familiar with the feeling of space between your fingers and the racket handle

FEEL this movement!
REMEMBER this
movement

If you now raise your hand very slightly up and down you will feel the racket lightly bouncing within your hand.



*Keep the loose feeling and bounce the racket **INSIDE** the hand*

This bounce needs to be repeated until it is easy for you to recognize. This is a subtle feeling and requires concentration and some repetition.

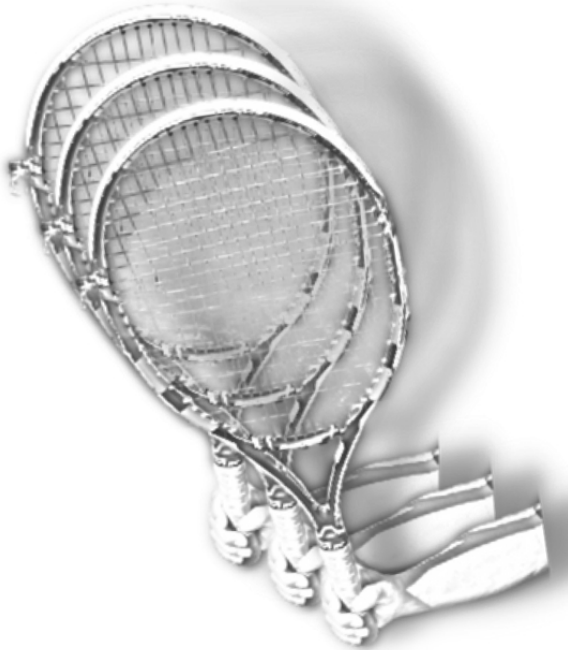
Find THE BOUNCE



When you next play, take the racket back and search for the feeling of the bounce of the racket in your hand before every forehand! Unless the ball is very fast or very deep then there is always time to feel the bounce of the racket inside your hand. When you feel the bounce of the racket, as it moves freely within the grip, then your hand and wrist will both be relaxed and loose.

You will now be feeling the racket with the same freedom that the best players in the world experience and utilize.

Feel THE BOUNCE



Whether the point is big or small. The moment that you feel the stress and tension of the situation affecting your grip of the racket. The moment that you feel the racket is like a foreign object in your hand. The moment that you feel the stroke lacks rhythm and shape.

Use THE BOUNCE

The bounce of the racket removes all of the problems associated with stress and tension in the hand and in the arm and will facilitate relaxed and fear-free tennis.

**Find the BOUNCE.
Feel the BOUNCE
Use the BOUNCE**

To see this lesson on YouTube

[https://www.youtube.com/watch?
v=C9O2GZkPcwU&list=PL0bA4JdHfGN78sdJWrw9B7__sEenkIIH
Z&index=12&t=176s](https://www.youtube.com/watch?v=C9O2GZkPcwU&list=PL0bA4JdHfGN78sdJWrw9B7__sEenkIIHZ&index=12&t=176s)

Service Leg Drive Biomechanics

The kinetic chain of the tennis serve involves the sequential movement of various body parts, starting from the ground up, in order to generate and transfer power to the ball.

The sequence of the kinetic chain in the lower body during the tennis serve is as follows:

- 1. The feet provide a stable base of support and help to transfer force into the ground.*
- 2. The force is transmitted up through the legs, activating the quadriceps muscles to extend the knee.*
- 3. The hips rotate and generate power, aided by the gluteal muscles.*
- 4. The trunk rotates and contributes to the power and speed of the serve.*
- 5. The shoulder, arm, and hand work together to deliver the ball with the desired speed and spin.*



Anatomically, the kinetic chain begins with the feet, which provide a stable base of support and help to transfer force into the ground.

In the feet, the main muscles involved in the serve include the tibialis anterior, which helps to lift the foot and ankle during the serve, and the gastrocnemius and soleus muscles, which work together to plantarflex the foot and provide power during the push-off phase.

The joints in the feet that are involved in the serve include the ankle joint, which allows for plantarflexion and dorsiflexion, and the metatarsophalangeal joint, which allows for toe extension.

The ankle joint is a hinge joint that allows for plantarflexion (pointing the toes down) and dorsiflexion (pointing the toes up). The gastrocnemius and soleus muscles in the calves work together to plantarflex the foot during the serve, providing power during the push-off phase.

From there, the force is transmitted up through the lower body, including the legs, hips, and core.

In the legs, the main muscles involved in the service include the quadriceps, which extend the knee during the service, and the hamstrings, which help to decelerate the leg during the serve. The joints in the legs that are involved in the service include the hip joint, which allows for a range of movement including abduction, adduction, flexion, and extension.



The hip joint is a ball-and-socket joint that allows for a wide range of movement, including abduction (moving the leg away from the midline of the body), adduction (moving the leg towards the midline of the body), flexion (bending the leg at the hip), and extension (straightening the leg at the hip). During the service, the hip joint is used to generate power and rotation in the lower body.

The knee joint is a hinge joint that allows for flexion (bending) and extension (straightening) of the leg. During the service, the quadriceps muscles contract to extend the knee, while the

hamstrings work to decelerate the leg during the follow-through phase of the serve.

Other muscles and joints in the lower body that are involved in the service include the gluteal muscles, which help to stabilize the hip and generate power, and the lumbar spine, which allows for rotation and extension.

The kinetic chain also involves the upper body, including the shoulder, arm, and hand. These body parts work together to generate the power and speed needed to hit a serve.

Biomechanically, the service involves a number of complex movements, including hip rotation, trunk rotation, shoulder abduction, external rotation, elbow extension, and wrist flexion. These movements work together to create the whip-like motion that is necessary to hit a powerful serve.

In addition to these anatomical and biomechanical factors, the service also involves proper technique and timing in order to be effective. This includes the use of proper grip, proper footwork, and proper body position at the moment of impact. The muscles and joints in the feet and legs play a crucial role in the tennis serve, as they provide the foundation for the rest of the kinetic chain.



Serve Leg Drive

One Minute Solution Solution - Don't bend your knees, the KNEES BEND

When watching the pro players, their service actions can seem like a very simple and easy movement. The stroke is wonderfully synchronized and has such rhythm. However, when we try to serve, the simplicity and the easy rhythm disappear and are replaced with a confusing and bewildering amount of things to think of.

This problem is very real, there are about 720 muscles in the human body and the service stroke involves about 700 of them. So, there is a lot happening in a very short space of time.

The secret to coordinating your service is to understand how kinetic chains work and then to apply this understanding to your stroke.

All kinetic chains work from the ground up.

This is an unwavering fundamental of how the human body functions.

So a correct and biomechanically sound service stroke also has to be created from the ground up.

Yet in traditional tennis coaching, many players are told to bend their knees.

The advice to bend the knees is counterproductive because when focusing on the knees, everything below the knees, the calf muscles, the Achilles, and the feet are all essentially locked and provide no assistance to the power of your service.

This means that power and energy can only begin from the knees up. But the knee is not designed like the ankle. The ankle and the feet will create momentum, the knees can only direct momentum. So concentrating on bending the knees results in less power and more effort.

One Minute Solution



Instead of thinking about bending your knees, you must think about raising your heels.

How high and when to begin raising your heels varies a lot from player to player. The most common point to begin the raising of your heels is when your racket reaches the height of your shoulder.

**When you focus on bending your knees the lower leg is essentially useless.
When you focus on raising your heels your the knees will bend.**



But now they are bending as part of a kinetic sequence that began from the ground and is working its way up. The

feet, ankles, calf muscles will all provide power and speed into your service.

Most people reading this will be sitting down.
You did not think about bending your knees to sit down.

The knees simply bent as part of a smooth harmonious kinetic chain.

Use this same smooth harmonious and natural kinetic chain to power your service.

Sometimes less thought creates better results.

When children are very young, they learn to walk and then they learn to talk.

Imagine a world where children learn to talk and then they learn to walk!

There would be experts on TV to advise you on how to teach your child to walk. There would be books entitled "Teach walking in 3 easy steps"
Other parents would be telling you how they taught their children to walk.

Yet, because we cannot talk to them, children all over the world learn to walk at about the same time in about the same way.

The human body and the human brain are amazing!

Use the same skills that enabled you to learn to walk and to sit in the chair you are in now to naturally develop your service into a powerful stroke.

By raising the heels you will instantly experience a smooth and pure transfer of energy from the feet, through the legs into the body, and out to the racket and the ball.

Don't bend your knees. Raise your heels, let the knees bend and your service will have more power today



To see this lesson on YouTube

[https://www.youtube.com/watch?v=Qg9Jb8C2jHU&
list=PL0bA4JdHfGN5iNYG5SY69Au1BDVfy_fA2
&index=12&t=215s](https://www.youtube.com/watch?v=Qg9Jb8C2jHU&list=PL0bA4JdHfGN5iNYG5SY69Au1BDVfy_fA2&index=12&t=215s)

STEPHEN JAMES BOURNE

I have been involved in tennis all of my life.

As a player I was a good hitter, I was ranked in the top 30 players in the UK, often helping professional players prepare for matches.

Having previously coached in the USA, Arabia and the UK I ran my own tennis academy for many years on the Eastern coast of Spain.

For over 20 years we worked with every imaginable type of player.

Beginner to professional, young and old (my oldest client was 93 years of age)

Aided by a large team of coaches I developed a unique approach to teaching (see The One Minute System on page 3)

This system proved to be amazingly successful at all levels.

During this time both recreational and competitive players requested online help and written information on the teaching methods that they had experienced and benefitted from.

The pandemic gave me the opportunity to take this teaching philosophy online and the One Minute Tennis YouTube channel began.

The channel enjoyed amazing success with over 2,000,000 views in the first year. <https://www.youtube.com/channel/UCbmBx0-ZS9V7KSbyhrZcYqw>

The purpose of this series of Tennis Solutions Books is to help recreational and developing players use the techniques of the best players in the world. The combination of the scientific principles of the latest techniques in tennis along with my quick and easy-to-follow One Minute Solutions make a complicated game simple and will help you to play better tennis today!



**GET YOUR
ONE MINUTE
TENNIS
SERVE SOLUTION
NOW**



